

	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering (PCCoE) (An Autonomous Institute) Affiliated to Savitribai Phule Pune University (SPPU) ISO 21001:2018 Certified by TUV	
Full Stack Development Laboratory		

Assignment-3

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Assignment 3

Title:

Design a Dynamic Web Application like Fashion Store / E-Commerce Product using HTML, CSS & JavaScript (Bootstrap CDN if required)

Objectives

- To understand the development of a dynamic web application using HTML, CSS, and JavaScript.
- To design an e-commerce style interface for displaying fashion products.
- To implement interactivity using JavaScript (product display, buttons, dynamic actions).
- To apply Bootstrap components for responsive and modern UI design.
- To learn client-side scripting for improving user experience.
- To simulate basic e-commerce functionalities such as product listing, add-to-cart interaction, and responsive layouts.

Outcomes

- Students will be able to create a dynamic web interface using JavaScript.
- Students will understand how front-end technologies work together in web application development.
- Students will gain experience designing responsive e-commerce layouts.
- Students will learn DOM manipulation and event handling using JavaScript.
- Students will be able to design user-friendly product pages similar to real online shopping platforms.

Theory Concept

A dynamic web application is a website that updates content automatically based on user interaction without reloading the entire page. In an e-commerce fashion store, dynamic behavior improves usability by allowing users to browse products, view details, and interact with items smoothly.

1. HTML (Structure Layer)

HTML provides the basic structure of the e-commerce website. It defines elements such as:

- Navigation bar
- Product cards
- Product sections
- Images
- Buttons
- Cart section

HTML acts as the skeleton of the application and organizes content systematically.

2. CSS (Presentation Layer)

CSS is used to style the application by controlling:

- Layout structure
- Colors and themes
- Fonts and typography
- Spacing and alignment
- Hover effects and transitions

CSS improves readability and enhances the overall shopping experience by making the interface visually attractive.

3. JavaScript (Behavior Layer)

JavaScript makes the website dynamic by adding interactivity and real-time updates.

Key functionalities implemented:

- Displaying products dynamically using JavaScript arrays
- Add-to-cart functionality
- Button click event handling
- Updating cart count without page reload
- DOM manipulation
- Simulating shopping cart behavior

JavaScript enables client-side scripting, making the application responsive and interactive without refreshing the webpage.

4. Bootstrap Framework

Bootstrap is a responsive front-end framework used through a CDN link. It provides ready-made components and a grid system for modern UI design.

Bootstrap features used in this project:

- Grid system for layout alignment
- Responsive navigation bar
- Cards for product display
- Buttons and forms
- Mobile-friendly responsiveness

Using Bootstrap reduces development time and ensures consistent and professional design across devices.

5. Dynamic Web Applications in E-Commerce

Modern fashion e-commerce platforms rely heavily on dynamic interfaces to improve customer engagement. Features such as:

- Dynamic product listing
- Cart updates

- Interactive buttons
- Responsive layouts

enhance user interaction and simulate real online shopping environments.

This experiment helps in understanding how front-end technologies combine to create practical web applications.

Dataset Used

No external dataset was used for this experiment.

The application uses manually created sample product data. The data is stored in JavaScript arrays and objects and dynamically rendered on the webpage using DOM manipulation techniques.

The product dataset includes:

- Product name
- Price
- Image
- Description
- Category

All data used is static and hard-coded within the JavaScript file for simulation purposes.

Index.html –

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Fashion Store</title>
```

```
  <!-- Bootstrap CDN -->
```

```
  <link
```

```
    href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
    rel="stylesheet">
```

```
  <!-- Custom CSS -->
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
  <!-- Navbar -->
```

```
  <nav class="navbar navbar-dark bg-dark">
```

```

<div class="container">
  <span class="navbar-brand mb-0 h1">Fashion Store</span>
  <button class="btn btn-warning">
    Cart (<span id="cart-count">0</span>)
  </button>
</div>
</nav>

<div class="container mt-4">
  <div class="row" id="product-list">
    <!-- Products will be loaded here dynamically -->
  </div>
</div>

<!-- Bootstrap JS CDN -->

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.m
in.js"></script>

<!-- Custom JS -->

<script src="script.js"></script>

</body>
</html>

```

Style.css –

```

body {
  background-color: #f8f9fa;
}

```

```
.card {  
  transition: transform 0.2s;  
}
```

```
.card:hover {  
  transform: scale(1.05);  
}
```

```
.card img { height:  
  250px;  
  object-fit: cover;  
}
```

```
.price {  
  font-weight: bold; color:  
  green;  
}
```

Script.js-

```
const products = [  
  {  
    id: 1,  
    name: "Men's Jacket", price:  
    2499,  
    image: "https://images.unsplash.com/photo-1520975922203- b5b3b7a3b5b6"  
  },
```

```
{
  id: 2,
  name: "Women's Dress", price:
  1999,
  image: "https://images.unsplash.com/photo-1520974735194-
0f88f0b6e78b"
},
{
  id: 3,
  name: "Sneakers",
  price: 2999,
  image: "https://images.unsplash.com/photo-1528701800489- 20be3c3c7e8a"
},
{
  id: 4,
  name: "Handbag", price:
  1499,
  image: "https://images.unsplash.com/photo-1512436991641- 6745cdb1723f"
}
];
```

```
let cartCount = 0;
```

```
function displayProducts() {
  const productList = document.getElementById("product-list");
```

```

products.forEach(product => {
  productList.innerHTML += `
    <div class="col-md-3 mb-4">
      <div class="card">
        
        <div class="card-body text-center">
          <h5 class="card-title">${product.name}</h5>
          <p class="price">₹ ${product.price}</p>
          <button class="btn btn-primary" onclick="addToCart()"> Add to
            Cart
          </button>
        </div>
      </div>
    </div>
  `;
});
}

```

```

function addToCart() {
  cartCount++;
  document.getElementById("cart-count").innerText = cartCount;
}

```

```

displayProducts();

```

Webpage –

